Multimedia Appendix for

iGrow: A Smart Agriculture Solution to Autonomous Greenhouse Control

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1. Supplementary Tables

Table S1. Basic state variables are defined in our model.

Category	Name	Description	Unit
Outside weather (w)	Iglob	outside solar radiation	W/m^2
	Tout	outside temperature	$°C$
	RHout	outside humidity	$\%$
	CO_2out	outside CO_2 concentration	ppm
	Windsp	outside wind speed	m/s
	Tsky	virtual sky temperature	$°C$
Indoor climate (c)	AirT	indoor temperature	$°C$
	AirRH	indoor humidity	$\%$
	AirCO_2	indoor CO_2 concentration	ppm
	PAR	photosynthetically active radiation	$umol/s$
Crop state (g)	LAI	leaf area index	-
	PlantLoad	number of growing fruits	$fruits/m^2$
	NetGrowth	photosynthesis net growth	$kgCH_2O/s$
Production (p)	FW	fruit fresh weight	kg/m^2

Table S2. Basic actions are defined in our model.

Action (setpoint)	Units
indoor temperature	$°C$
indoor CO_2 concentration	ppm
illumination on/off	-
irrigation on/off	-

2. Supplementary Figures

2.1 Deployment Facilities for Pilot Projects in the Real Autonomous Greenhouses



Figure S1. A real greenhouse managed by iGrow solution.

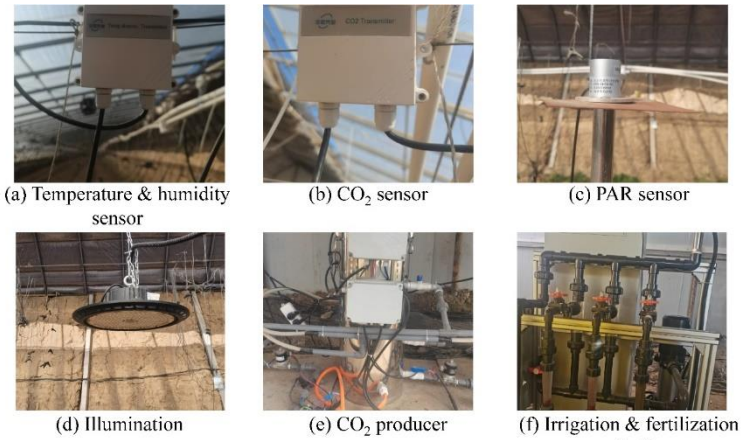


Figure S2. Some equipment of the real greenhouse.



Figure S3. Visualization platform. Shows the environmental monitor situation in different greenhouse of the 2nd pilot project. (From 2020-06-07 to 2020-06-14)

2.2 Experiment results of the 2nd Pilot Project in Real Autonomous Greenhouses

We plot the pair relationship among four states (See complete results from Supplementary Figure S4 to S13): AirT, AirCO2, AirPAR, and AirRH, which are the controlled variables by planting experts and the iGrow platform. Taking Supplementary Figure S10 as an example, we observe some useful insights:

(1) The greenhouse controlled by the control group exceeds 30℃ more frequently, which is not suitable for crop growth, sometimes even harmful to crop health;

(2) Between 17-27℃, photosynthetic enzymes are more active. In the meantime, the greenhouse controlled by the experimental group shows higher CO2 concentration and PAR, contributing to enhancing photosynthetic intensity;

(3) The greenhouse controlled by the control group has more humidity above 70%, which is harmful to crop health.

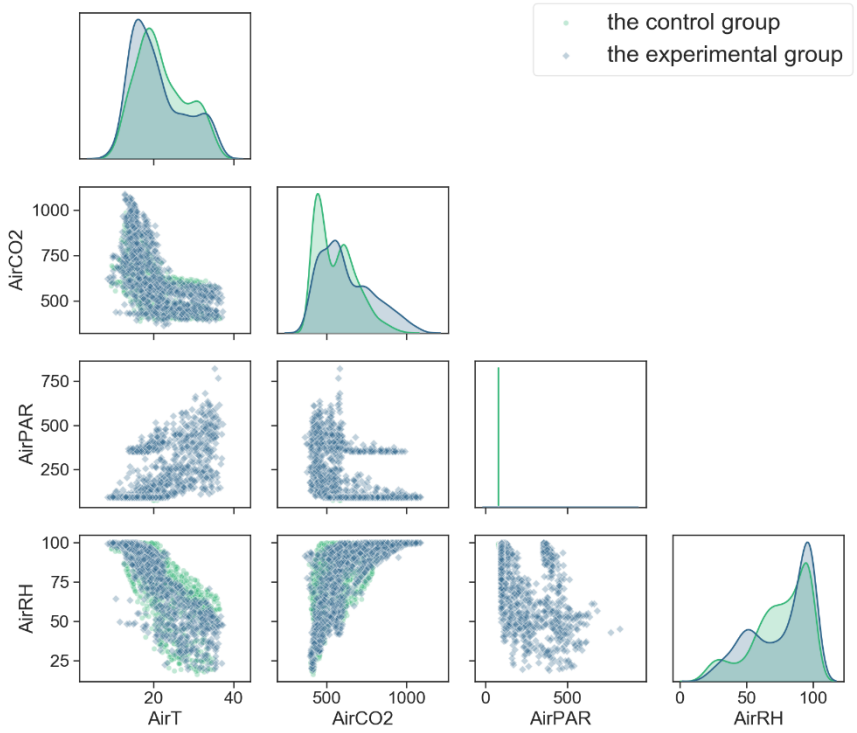


Figure S4. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh01 vs. the experimental group gh03).

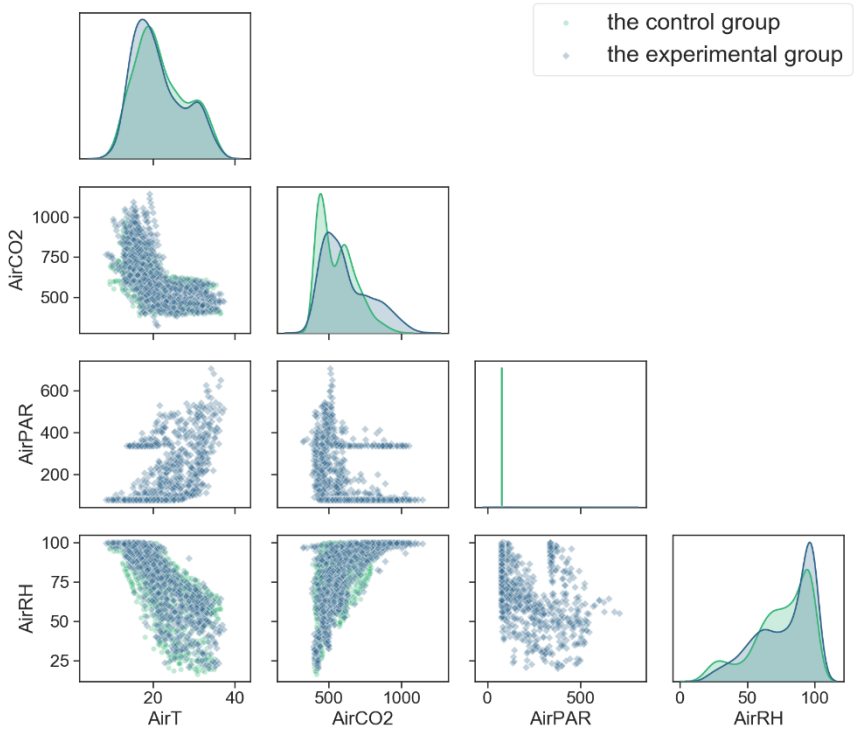


Figure S5. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh01 vs. the experimental group gh04).

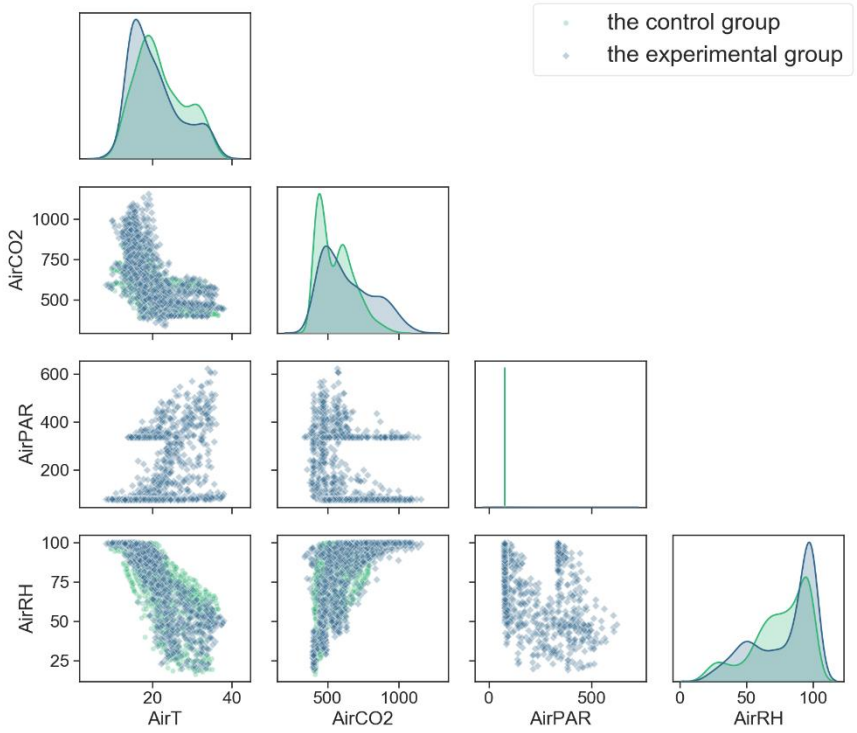


Figure S6. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh01 vs. the experimental group gh05).

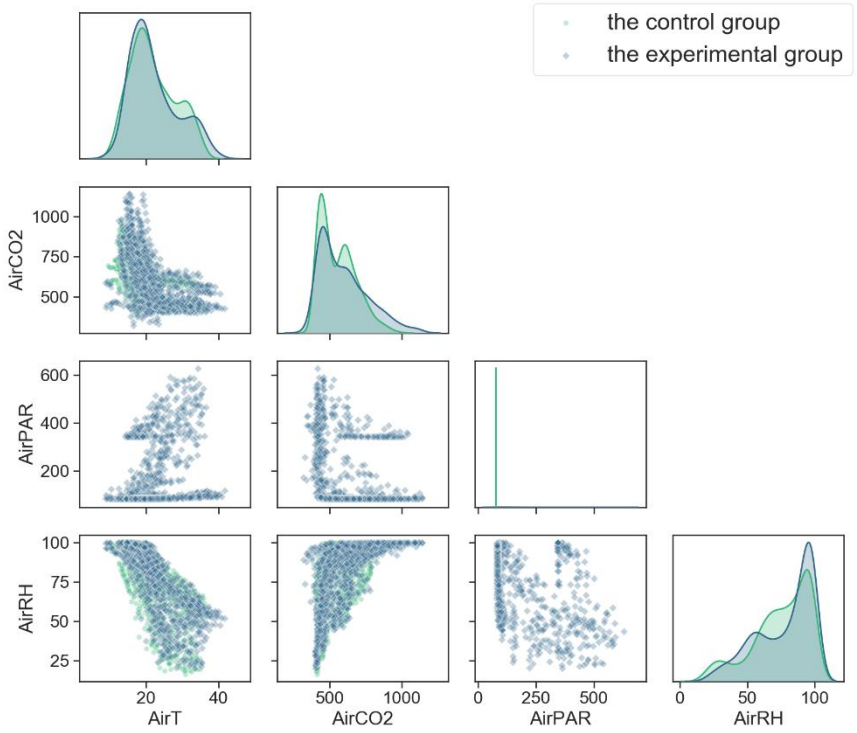


Figure S7. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh01 vs. the experimental group gh06).

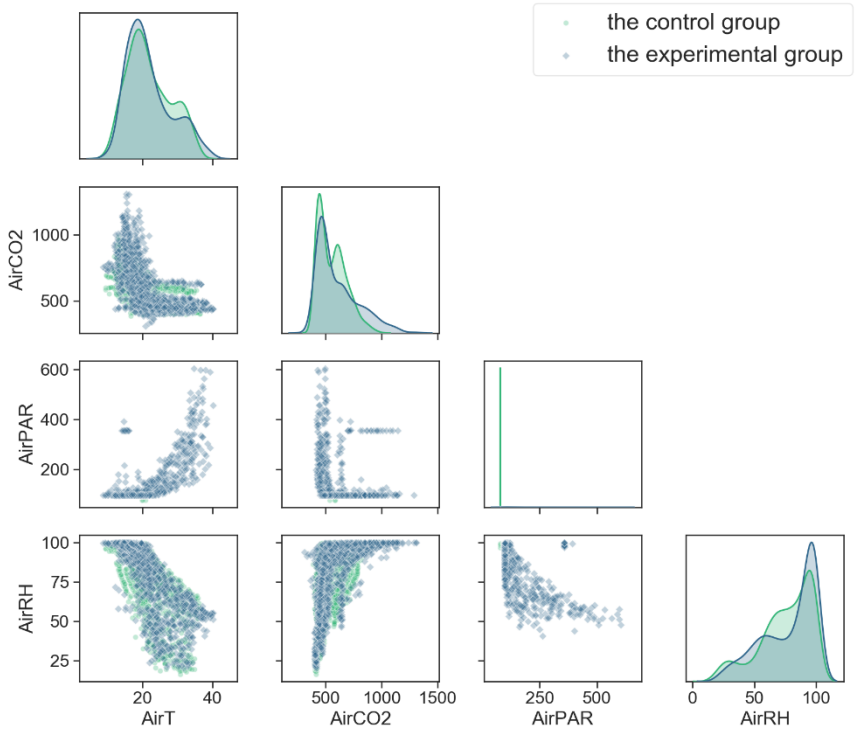


Figure S8. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh01 vs. the experimental group gh07).

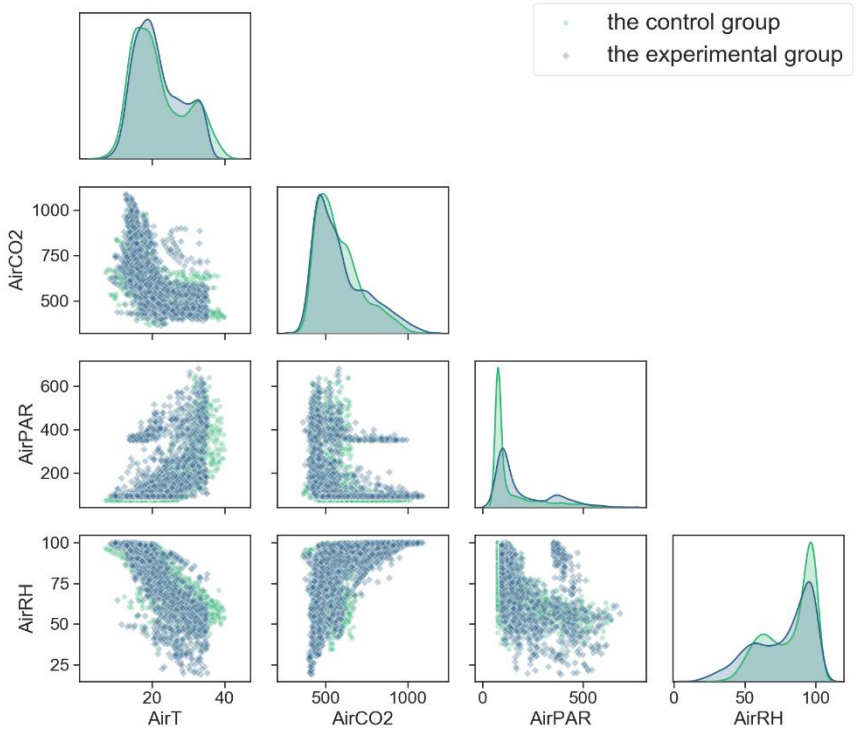


Figure S9. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh02 vs. the experimental group gh03).

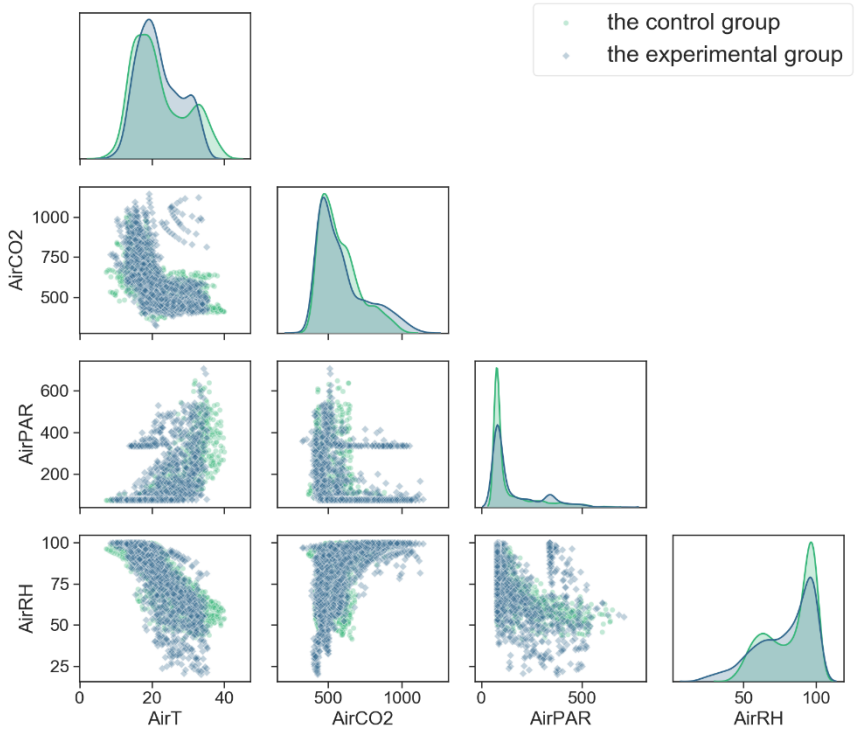


Figure S10. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh02 vs. the experimental group gh04).

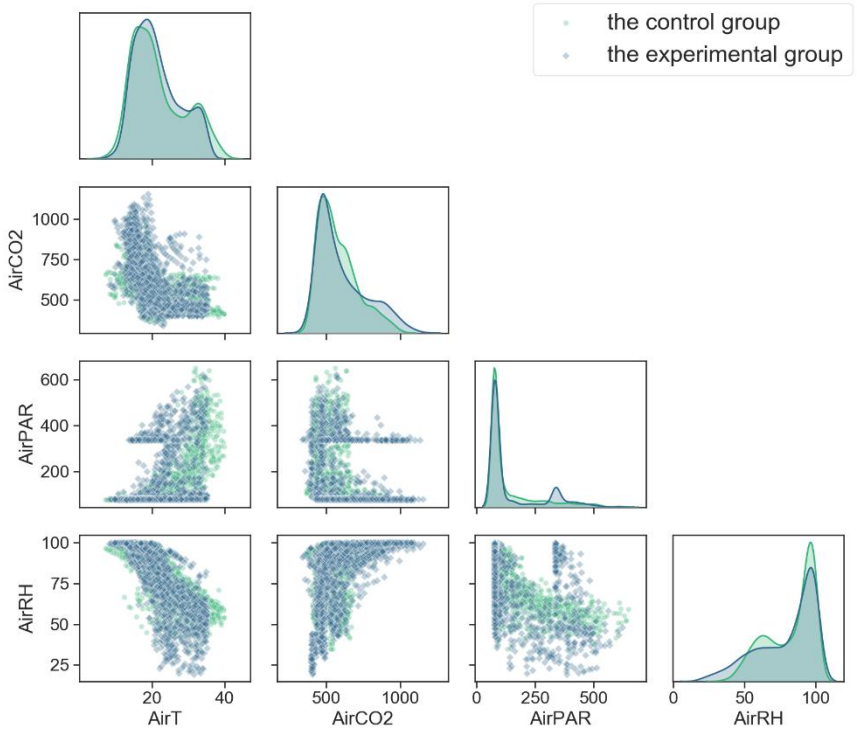


Figure S11. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh02 vs. the experimental group gh05).

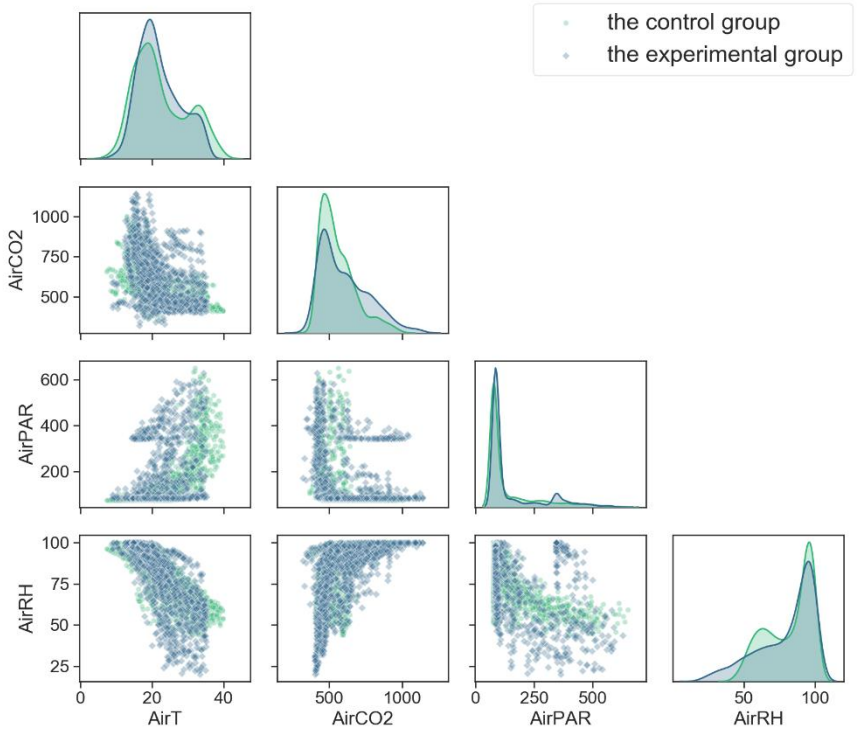


Figure S12. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh02 vs. the experimental group gh06).

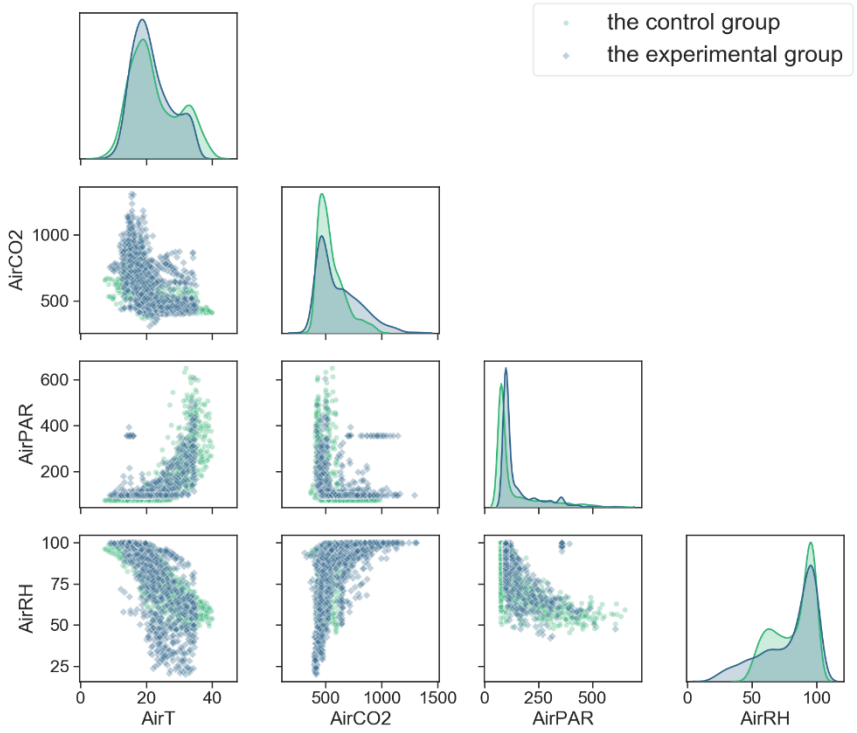


Figure S13. The relationship among AirT, AirCO2, AirPAR, and AirRH under the control strategies (the control group gh02 vs. the experimental group gh07).